

SECURITY CONSIDERATION FOR THE IPV6 BASED AIR TO GROUND SAFETY SERVICE COMMUNICATION

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Abstract

Currently European Airspace is using the ATN/OSI and ACARS protocols for the air to ground communication and the US is using ACARS protocol. There is a big movement in the US national airspace and European airspace to move the airplane air to ground safety service communication to the Internet based protocol, using Internet Protocol Version 6 (IPv6) from exiting ACARS and ATN/OSI. When the airplane moves to the IPv6 based air to ground communication, the airplane is essentially a mobile node equipped with multiple air to ground radio technology interfaces with several network interfaces. These network interfaces have different access technologies, both terrestrial and satellite communication (VHF, HFR, SATCOM etc.). Different requirements of different applications can result in different preferences of the interface or air to ground communication means that expose different attack surfaces, vulnerabilities and security issues. Network connections should be placed in the best possible interface and the security related to the various access networks should be considered to protect both data and the control of the plane traffic. Due to multiple technologies and multiple access network technologies, the Aviation safety service IPv6 air to ground network needs to address multiple security risks.

In this paper we present an overview of security issues and explore the challenges and motivations of security measures for the multiple access networks to protect the safety service IPv6 air to ground communication. This review is critical because securing future air to ground communication for the safety service is a difficult problem. It can be an obstacle to aviation IPv6 deployment in the mission-critical environment. We also present the architectural goals and security measures that should be considered while deploying the IPv6 for the safety service air to ground communication.

Introduction

The traditional Aircraft Communications Addressing and Reporting System (ACARS) has been used in air/ground communication between aircraft and ground entities for well over 30 years. The airline industry first used ACARS in 1978 to carry Aeronautical Operational Control (AOC) messages, specifically to assist with the reporting of aircraft OOOI (Out of the gate, Off the ground, On the ground, and Into the gate) events [1]. Since the early days, the application of ACARS-based messaging has increased rapidly in size and quantity. Currently ACARS is used worldwide for aircraft position and status reporting to airline back offices and airline operational management, as well as to Air Traffic Control (ATC). Europe has deployed an Aeronautical Telecommunication Network (ATN) using the OSI protocol (ATN/OSI) for ATC use. Both communication technologies (ACARS and ATN/OSI) are well-known protocols offering limited security, however, the lack of internet connectivity, the single-purpose protocols, and the specialized communication media, perhaps make ACARS and ATN/OSI less likely to attract bad actors.

There are various industry and regulatory activities (AEEC IPS Working Group, United States RTCA SC-223, European EUROCAE WG-108, and ICAO WG-I) under way to develop aviation standards to migrate air/ground safety service communications from ACARS and ATN/OSI to the Internet Protocol Suite based aviation safety service ATN/IPS. Internet Protocol Suite IPv6 is the selected protocol for the air/ground data communications to become an alternative to ACARS and ATN/OSI. It is expected that IPv6 air/ground communication exposes the aviation safety services data communication network to various cyber security threats. Bad actors could attempt attacks on aircraft systems using ATN/IPS. Once ATN/IPS is widely used, the consequences of an advanced persistent threat would be dire. The ATN/IPS network could become congested, airline operations could be affected, and safety critical systems in the aircraft and on the ground could be affected by threats using the

IPS directly or indirectly. As shown in Figure 1, the IPS aircraft is a multi-homed system housing multiple interfaces with different access networks and service providers, including in the aircraft and on the ground systems. There is a need for strong security to protect the aircraft within each access network and between the aircraft and each service provider.

When the air/ground communication uses an internet protocol based network, as shown in Figure 1, it can be a large mobile IP network, since a node (the airplane) moves across multiple continents, and each continent will have a large number of airplanes. Because of this global scope, security becomes a tremendously important issue. Seamless mobility support is an essential requirement of the ATN/IPS and a desirable characteristic for the applications using the network. In mobile ATN/IPS networks, mobile nodes join or leave the networks frequently and the network topology constantly changes, which makes network management a challenge. In the mobile environment, where the network topology is constantly changing, multipath routing is vulnerable. An IP based network that lacks integrity and security checks may face the challenge of dealing with malicious actors, who fabricate packets in order to disrupt communications [2], [3]. Without security and integrity, all users would be potentially vulnerable to misuse of the services provided using ATN/IPS. Hence, it is necessary to deny access to compromised, defective, and illegitimate nodes. This requires mutual authentication between a mobile node (the airplane) and the network.

The end to end communications for the existing air/ground system and the context of how IPS is incorporated is depicted in Figure 1. Many current avionics designs do not seamlessly support application security. Historically, stand-alone avionics systems were proprietary implementations. A large number of independent legacy avionics systems co-exist across the aviation network and they may not provide any security functions. Their silo nature conflicts with the open spirit of the internet protocol; therefore introduces problems of interoperability and service-level interaction, which limits the benefits of internet protocol based systems and could impede large-scale IPS deployment.

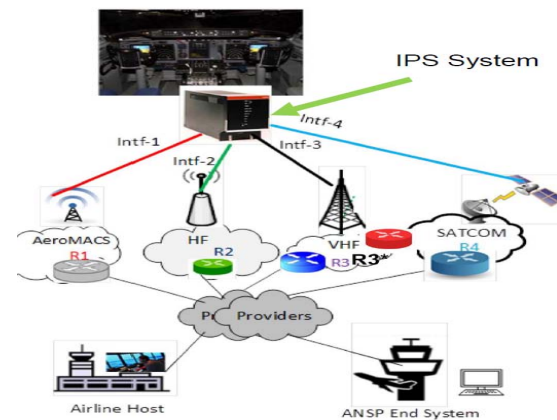


Figure 1. IPS Communication Entities {Rn = Routers, Intf-n = Interface}

In this paper, our intent is to explore the security considerations for the Aviation safety service IPv6 air/ground communication.

Related Works

A vast amount of related literature can be found on the topic of network security in general [4], [5], [6], [7], security for wireless networks, [8] and [9] overall network security. The literature enabled an assessment of the approach to take for the proposed air/ground IPS approach. Hansman et al present a four dimensional approach to attack taxonomy, which includes target, attack vector, exploit, and vulnerabilities [10]. Basin et al [11], and [12] present the security vulnerabilities in the embedded system in various protocol layers. Buttyan, Mirjalili et al [13], [14] present the various security threats related to embedded systems. As an excellent entry point, sources [15] and [16] provide an extensive overview on the security needs of IP Network communication. Three aspects, mutual authentication, data integrity and privacy, will be examined in more detail, as they are crucial with regard to the security requirements. Other aspects, such as secure routing are also relevant for the anticipated application scenarios.

There has been significant progress in the security architecture on embedded ECC public key cryptosystems [17], [18]. Embedded internet ECC public key cryptosystem use is described in the [19] paper. In source [20], one of the main criteria for session key agreement is provision of forward secrecy. The forward secrecy means that disclosure of long-term secret keys does not lead to exposure of

session keys from earlier runs. Mutual authentication is described in the Kizza et al [21] scheme. In their scheme, when mutual authentication is realized between the user and the server, they agree on a session key to be used for protecting their subsequent communications. There has been extensive research done on mitigating the effect of IP spoofing. A similar approach was proposed by Duan et al [22], introducing the concept of the feasible upstream router and deeming any packet that does not originate from such router, a spoofed packet. Wang et al [23] presents the secure routing protocol provision. There are more formalized approaches and these provisions should be considered in data communication security. This includes mutual authentication, data integrity, non-repudiation and privacy [24], [25] and [26].

Astatke et al [26] describes the method to use public key cryptography in wireless sensor networks using Elliptic curve cryptography (ECC) as the public key cryptography scheme [27] and Advanced Encryption System (AES). ATN/IPS requires an efficient way to enable end to end secure communications. ECC cipher suites in the RFC [28] such as cipher suite TLS_ECDHE_ECDSA_WITH_AES_256_CBC_SHA384, provides security that is considered reasonably future proof. ECC is used to generate key pairs, and provide entity authentication, message integrity and privacy. Ephemeral Elliptic Curve Diffie-Hellman (ECDHE) algorithms provide the forward secrecy.

ATN/IPS: Systemigrams

Here we will provide short relationships between all parts using the systemigrams [29]. Figure 2 illustrates the systemigrams for the ATN/IPS events in order to understand the interconnected operational environment to include security measures, air to ground and ground to ground network operation, risk, and government regulatory framework, etc., across the entire aircraft to ground end to end communication. Figure 2 starts with the Airborne ATN/IPS (the system of interest) and ends with the outcome of the system of interest by exchanging secure messages to/from aircraft and ground systems. To send and receive messages, the aircraft uses VHF, HFR, and SATCOM radios links. The ground network uses the access network to relay messages to or from the aircraft to the aviation safety service core network. The core network routes

messages between the aircraft and ground systems, such as ANSPs, Airline Operational Centers (AOCs), vendors, OEMs, and engine manufacturers. The aviation core network and access networks are managed and operated by the service providers. Government regulators are responsible for setting the regulatory guidance for the certification of the airborne system, network end to end latency, performance, and safety and security requirements. Airlines, ANSPs and others pay for network usage, which is the income for the network service providers. Network providers invest in the core network to meet communication latency, safety and security objectives. Figure 2 also represents the scope of the ATN/IPS. While the scope does not include the goal of the ATN/IPS, it does include nodes for that purpose with the addition of nodes representing most elements.

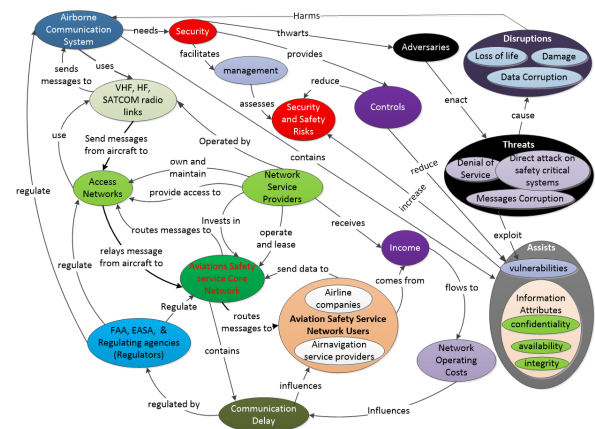


Figure 2. ATN/IPS Systemigrams

Security Threats

Threats in the ATN/IPS context can be any person, machine or event that can cause damage of data, ground or avionics systems. Since threats can also be natural (wind, lightning, fire) or accidental (accidentally data loading or removing content from avionics), these kinds of threats are not considered here. We briefly provide the summary of significant threats in this section.

Disclosure of Network Traffic or Data: This may include disclosure of airline data that may damage confidentiality. Unauthorized release of airline data may hurt or damage an airline reputation. For the ATC data, data disclosure may or may not be considered a threat.

Data Modification: Illicit modification of data is an attack on data integrity. Modifying data, avionics and/or ground system software can create significant risk. If the airplane flight profile data is changed and sent to the airplane, the pilot may change airplane speed, altitude, or heading profile, placing the airplane in a harmful position. This may introduce a hazardous condition.

Disruption of Aviation Network: ATN/IPS provides the communication means for airline operators, vendors and air traffic control to exchange messages in a timely manner. If network disruption occurs, it will not provide the needed functionality on time. The disruption of the network can affect operation and safety of the airplane, its crew and passengers. Several conditions may lead to network disruption including vulnerability, exploitation, a network with single point of failure, improper maintenance of network or equipment, unauthorized access to network and manipulation of network control traffic.

Common attack types threatening air-ground communication in particular and the explanations are shown in Table 1.

Table 1. List of Common Threats

Threat (Adversary Goal)	Description
Black Hole	An adversary gets all air-ground traffic in the network or directed to a certain node to flow by advertising false routing information. Then it can simply drop packets it was supposed to forward to the airplane or air traffic control or airline. It is one of the critical air-ground network attacks in which the adversary directs the communication from some victim node to the adversary node. The adversary node accepts the data packets, but does not forward to the neighboring nodes.
Denial of Service (DoS)	DoS attack means preventing authorized users from

Threat (Adversary Goal)	Description
	accessing services offered by the network. An even more severe form of the DoS attack is the distributed DoS (DDoS) attack, where a group of adversaries work together in preventing authorized users from accessing the network services.
Eavesdropping	Eavesdropping is a passive attack. An adversary listens to the network and routing traffic and attempts to discover the nodes information. Detection of passive attacks in wireless networks is usually difficult but network nodes are generally able to protect their data by encryption and thus prevent eavesdroppers from getting any valuable information.
Impersonation, Man-in-the-Middle	One type of impersonation attack is a man-in-the-middle attack, where an adversary reads and possibly modifies messages between the airplane and ground nodes, so that neither of the end nodes knows that they have been attacked.
Wormhole	Wormhole attack means that an adversary selectively tunnels packets from one place in the network to another. Wormhole is the tunnel between the two adversaries, which are linked via a private network connection. Due to airplane shared broadcast radio channel, the attacker can create a wormhole for packets addressed to any node in the network.

Threat (Adversary Goal)	Description
Resource Consumption	In this threat an attacker tries to wear out the network's resources and/or the nodes in it. These are unnecessary requests for routes, injecting a large number of unnecessary packets into the network or forwarding unnecessary packets to nodes. Avionics nodes in general have limited resources of bandwidth and computational power. Power exhaustion attacks in avionics are a real, powerful threat because a node can no longer function in the network once the resources are starved.
Port Redirection	This is the trust exploitation attack in which an adversary finds a way around the security mechanism. If any system in the access network, core network or end user network is compromised by the attacker, then the attacker can exploit or devise a malicious mechanism that can redirect the traffic towards the adversary or directly to some undesired end point. The attacker may create a tunnel for communication and bypass the security barrier.
Port Scan or Sweep	For the aviation safety service network, port scan and ping sweep are common network probes typically used to run various tests against avionics or any other nodes to find vulnerable services. In port scanning the attacker sends a packet to each target port and the reply message indicates that either

Threat (Adversary Goal)	Description
	the port is open or closed, which is further helpful to launch an attack against a specific service.

Threat Analysis

The aviation network is vulnerable to multipath attacks, due to multiple air/ground links. The air/ground communication assets with potential attack paths and threats are shown in Figure 4. Attackers can flood an airborne or ground radio with data, by launching Denial of Service (DoS) attacks or corrupt modified messages [30]. In DoS attacks, intruders send a flood of packets preventing the system from performing normal communication.

Avionics air-ground communication has four basic types of important control messages: Router Solicitation - Route Requests (RS), Router Advertisement - Route Replies (RA), Route Reply Acknowledgement (R-ACK) and the last message consists of aircraft change of link/traffic preference and link status notification to the ground. The aircraft utilizes these messages to maintain the local connectivity of a node and notify the ground regarding changes of aircraft preferences and connectivity status. Avionics have a stateless protocol where the node updates its links preference and next hop reachability information every time it determine changes in air/ground link status.

Main security threats are interruption, interception, modification and fabrication of air-ground messages. In the last decade, the research emphasis has changed to the security issues. Security concerns are still far from solved. The attack types threatening air-ground communication in particular, and their explanations, are shown in Table 1.

Security Vulnerabilities

ATN/IPS vulnerabilities consist of weakness in any entity in the ATN/IPS network that can be exploited by a threat. It is essential to protect all air to ground ATN/IPS communication elements and devices from a malicious adversary. Doing so may reduce the chances to exploit the vulnerabilities.

As shown in the Figure 3 [31], which is based on the statistics from the reports of Computer Emergency Response Team/Coordination Center data, the number of exploited vulnerabilities has increased significantly last few years.

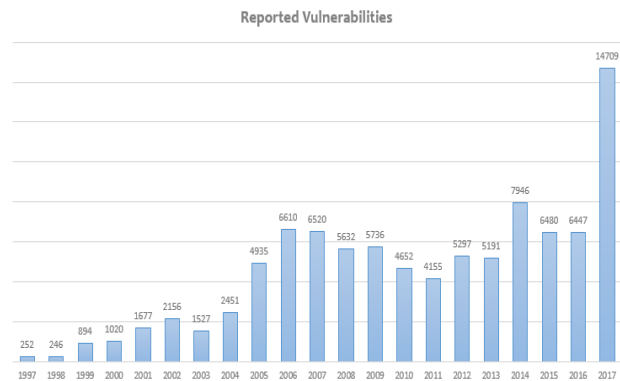


Figure 3. Number of Reported Vulnerabilities

As shown in Figure 4 [32], application programming interface (API) vulnerability is historically increasing. The ATN/IPS network must have a protection approach against vulnerabilities and include security measures in the avionics certification process.

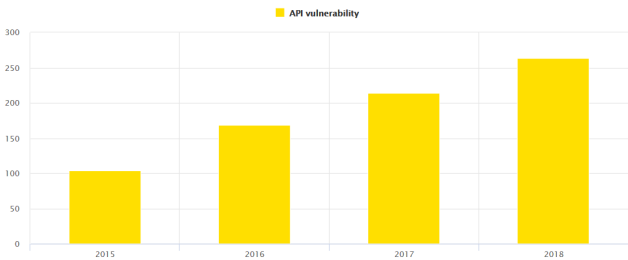


Figure 4. API Vulnerabilities

Avionics Certification Cycle

Experts may agree that today’s major challenge is not the security technology itself, but how to make appropriate procedures and controls for achieving secure ATN/IPS (especially avionics security). Attackers will always remain and possibly even grow in numbers. Emerging trends in the networked and connected arena are leading the world to a point where avionics will do even more automation and control than they are doing now. Technology development never stands still. Attacking tools will continue to advance, as will security solutions. If

avionics tries to stay up-to-date with every new threat, we must have a way to deploy security fixes on the avionics in a timely manner. Industry and regulators must look into the process and figure out how to fix major vulnerabilities, ensuring certified avionics stay protected. The current time consuming process will not be sufficient to address the evolving threats and changing security dynamics. Table 2 provides the airborne system identified threats, which are categorized and ranked.

Table 2. Threats and Associated Attack Goals

#	Threat	Adversary Goal
1	Corrupt Safety Critical Software or Systems	Harm airplane, harm people’s lives
2	Tampering or Corruption (a loss of integrity)	Impersonate controller or airline back office user
3	Sabotage	Loss of system availability, reduce system availability
4	Information Leakage of Disclosure (loss of confidentiality)	Steal information
5	Transaction Fraud	Miss-route data traffic
6	Harm the User	Break the Airborne Systems

The attack goals identified in Table 2 are modelled using attack trees.

Attack Trees

As an example, Figure 5 is the attack tree for attack Goal 1 of Table 2 (Harm airplane, harm people’s lives). On the left side tree branches the adversary sends specially crafted messages to airborne systems. To achieve this goal the adversary compromised the ground network, controller systems or airline systems as shown in the left side tree branches. An adversary can compromise the ground network systems with the ability to logon remotely and obtain the root access of the network system to change the router configuration. This can be achieved either by hacking the firewall, social engineering the firewall for rule changes, or exploiting the system

vulnerability. An adversary can also send specially crafted messages to the aircraft by accessing the controller or airline systems. On the right side tree branches, an adversary can modify the airborne system without authentication either by exploiting the airborne system vulnerability or accessing the system.

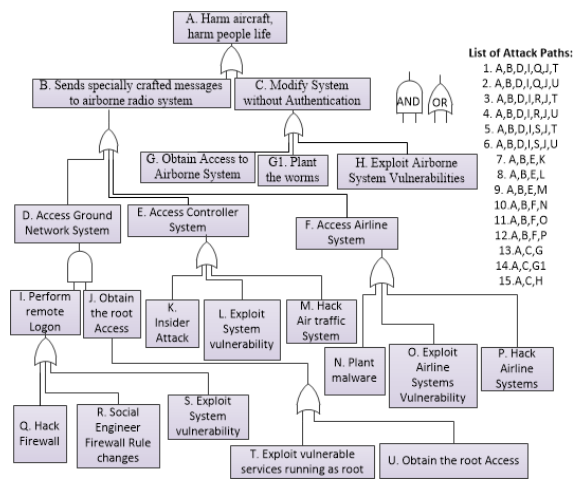


Figure 5. Attack Tree

Security Requirements

To prevent an attack or reduce the risk, preventive controls and security requirements can be planned and implemented for each activity contributing to an attack path in Figure 5 as listed below in Table 3.

Table 3. Security Requirements

Attack Leaf	Adversary Activity	Security Requirements
1-G	Obtain Access to Airborne System	Physical security, Fail safe, Cryptographic key generation, Cryptographic operation, Complete access control, Security attribute based access control, Integrity monitoring
1-H	Exploit Airborne System	Diversity, Depth, Record compromises

	Vulnerabilities	
1-N, 1-G1	Plant Malware or Plant Worm	Defense in Depth, physical, Fail safe, Complete access control, Security attribute based access control, Integrity monitoring, Unforgeable authentication
1-O	Exploit Airline Systems Vulnerability	Diversity, Depth, Record compromises, Defense in Depth, Physical, Fail safe, Complete access control, Security attribute based access control, Integrity monitoring, Unforgeable authentication
1-P	Hack Airline Systems	Same as previous row
1-Q	Hack Firewall	Diversity, Depth, Record compromises, User identification before any action, Complete access control, Security attribute based access control, Integrity monitoring, User identification before any action
1-R	Social Engineer Firewall Rule Changes	Diversity, Depth, Record compromises
1-S	Exploit System Vulnerability	Same as row 1-O

1-T	Exploit Vulnerable Service Running as Root	Diversity, Depth, Record compromises
1-U	Obtain Root Access	Diversity, Depth, Record compromises
Attack Leaf	Adversary Activity	Mutual authentication, Integrity, Availability

Concept of Operations

Air to ground communication threats may become more prominent, because adversary can potentially change mission critical messages or broadcast a wrong message to the airplane. Security is an important factor; mutual authentication between communicating entities, integrity and availability are main security requirements. Security measure also should properly sanitize specially crafted messages. The following sections provide the three possible security alternatives to provide safeguards for the ATN/IPS.

Option 1: Security Barrier in Each Airborne End Systems and Airborne Communication System and Each Safety Critical System

This option of security barrier involves putting the security perimeter in every airborne system. Each system provides security and optimum defense in the depth, fail safe, segregation of data flow and diversity. Each system checks the data entering the system and only allows types of data that are authorized to pass to the next systems. For example: access network offers security, the radios will provide the security for incoming and outgoing data, independent of other systems. The communication system offer security, so does every safety critical system, ATC application systems and airline back office application systems. This option prevents unauthorized data traffic and malicious data from entering the aviation safety service network. Each access control point where data and control traffic can be screened and, if necessary, inspected before passed through the barrier.

Option 2: Put Security Systems in Airborne Radios and Airborne Communication System

This option involves putting the security function in each airborne radios and communication system. This approach offers two independent barriers for the defense in depth, fail safe, segregation of data flow and diversity; however no security function is implemented beyond the communication system. Each radio system checks the data entering the system and only allows types of authorized data to pass data to the communication system. The communication system can further inspect the data traffic before passing through the barrier to other systems.

Option 3: Put Two Security Barriers in Airborne Communication System

This third alternative puts the security function and two barriers in the communication system. The security protection on every system is both highly complex and costly, especially with respect to the certification and technical security measures for the avionics. It is often the case that solutions have to be specifically developed and implemented to meet the needs of aviation, which means that off-the-shelf solutions are neither suitable nor readily available. Consequently, already costly security measures become even more expensive. Projects concerning security measures of new avionics and the upgrading of existing avionics are associated with very high costs. Due to the high costs, operators are often unwilling to accept such plans and may even oppose or reject them.

Security Return on Investment

Measuring the security risk is important for the business and investment justification. Slayton [33] describes the importance of measuring the security risk and associated security matrices. The aviation security framework must provide a cost-effective approach for security measures and controls. It will help avionics vendors, OEMs, suppliers, and operators of aviation safety service communication assess and manage cyber security risks.

Added security measures must be assessed to decide if it is worth the investment. Trade-offs between risks and benefits must be performed. If the risk is very small, it may not justify the addition of a security measure. In other cases the risk may be high and the system so complex that it becomes cost

prohibitive to develop, certify, deploy and maintain with the addition of security measures.

Verification and Validation

The ATN/IPS avionics require airworthiness certification (safety of flight) so that verification and validation follow airworthiness processes, which demands that all requirements are validated and tested. Tests are configuration controlled and peer reviewed. Requirements are validated in the early development phase with stakeholders. During the verification and validation, it is imperative to ensure that data and components of the airborne communication system are managed and software modified, created, or deleted by authorized personnel. Metrics are established to measure integrity by considering validation checks for data flows and input from different sources. Validation includes process and criteria validity for each message construct, which should be incorporated in the tests. It also requires that security compliance be maintained and monitored through the life cycle of the system.

Certification for security measures and certification framework for the vulnerability patch updates and their associated deployment are necessary to sustain the continuous security of the ATN/IPS.

Conclusion

This paper documents multiple aspects and security approach considerations for the ATN/IPS. Its purpose is to raise awareness for the ATN/IPS security engineering analysis. The use of attack trees can be slightly challenging to capture for ATN/IPS, because its characteristics of a complex system and multiple attack surfaces that are available. Attack trees can be difficult to analyze solutions against the tree. A problem with attack trees is, they capture attacks that are sequences of events and not appropriate for attacks that involve concurrent actions. To secure ATN/IPS, increased human intervention and increased cross-functional dependencies are required amongst the airplane communication system, access network, airline, ANSPs, regulatory/certification authorities, airplane manufacturer and avionics vendors. In the meantime the adversary may shift the attack and decision points

in a different direction and operate faster than the defender, thereby gaining the upper hand.

Based on the limited security assessment of the airborne communication system, it becomes obvious that a thorough and systemic approach to security will be time consuming, it requires to think like an adversary, and will be valuable to assess and understand what vulnerabilities in the system exist. Security measures can be put in to place to safeguard the critical safety infrastructure of the airplane, but they could become an easy target in comparison to IT systems with a faster safeguard deployment.

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